UNIT III

Current drones, use cases and business value

A 60-minute interactive session on topics 16–18

Power it, sense it, pay for it

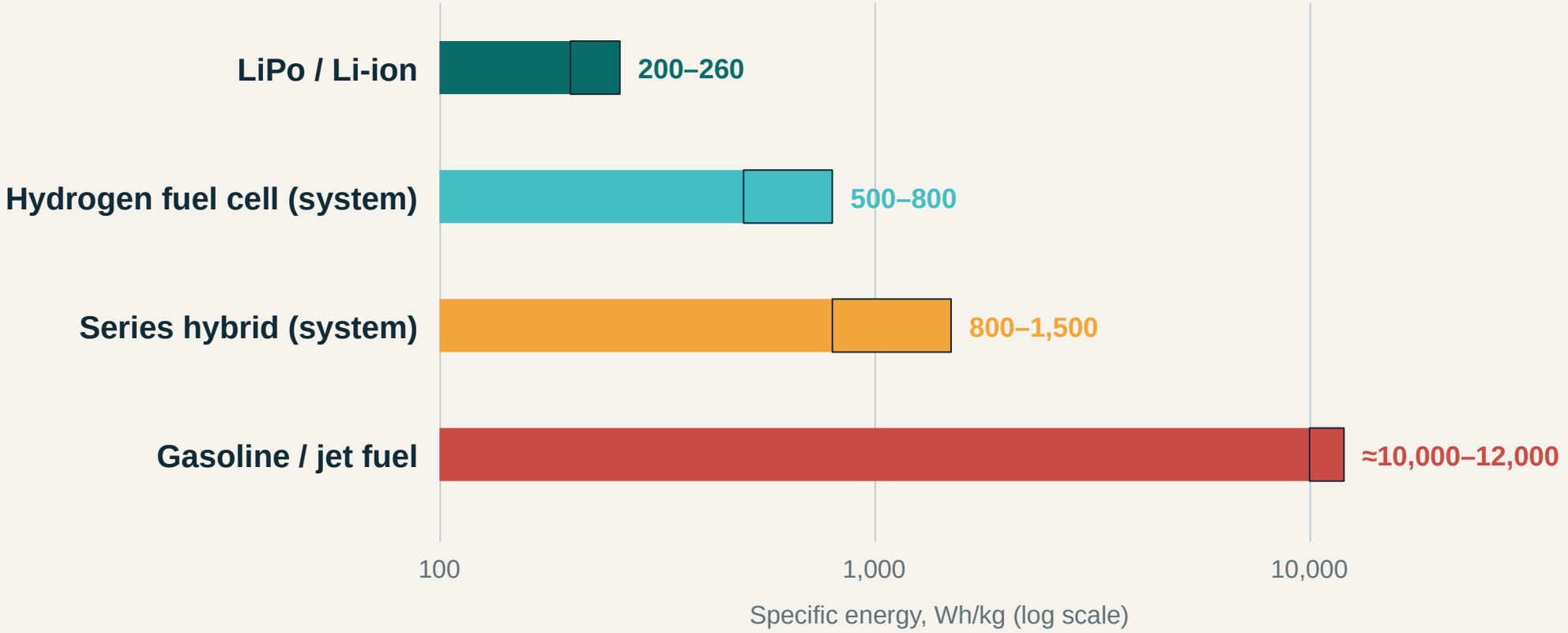
Today’s 60-minute plan

0–12	12–22	22–32	32–42	42–54	54–60
Energy and propulsion	Rotors and prop scaling	Payloads and positioning	Use-case matching	Value, ROI and payback	Exit ticket

Learning outcome

Choose a power source, payload and positioning method for a mission, and show the business case in numbers.

Fuel stores about 50× more energy per kilogram than a LiPo



But: a LiPo motor chain turns 75–85% of that energy into shaft power; a small engine manages about 20–30%. Electric still wins below roughly an hour of flight.

No propulsion wins everything; pick on the mission

Architecture	Specific energy	Efficiency	Vibration	Endurance	Typical use
Pure electric	200–260 Wh/kg	75–85%	Ultra-low	20–60 min	Mapping, cinema, inspection
4-stroke ICE	10,000–12,000	25–32%	Moderate	6–24 h	MALE fixed-wing
Series hybrid	800–1,500 (sys)	20–26%	Moderate	2–5 h	Heavy-lift multirotor, VTOL
H2 fuel cell	500–800 (sys)	45–55%	Zero	2–4 h	Long linear assets
Micro-turbine	11,500–12,000	10–16%	Low	15–60 min	High-speed targets

Rule of thumb: multirotors need fast thrust changes (under 10 ms) for attitude control, so engines usually drive a generator (series hybrid) rather than the rotors directly.

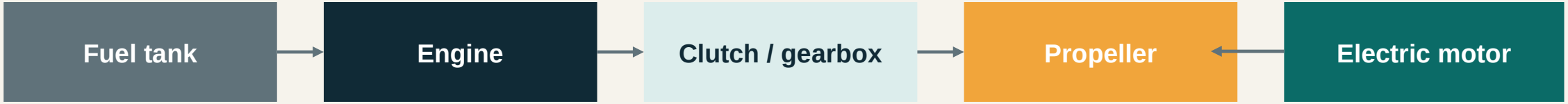
PANEL: CIRCLE THE DEAL-BREAKER IN EACH ROW

Hybrids: the engine either makes electricity or turns the prop

Series hybrid



Parallel hybrid



- **Series:** engine runs at its best rpm; battery covers thrust spikes. Common on heavy-lift multirotors.
- **Parallel:** fewer conversion losses at cruise; motor boosts take-off. Suits fixed-wing and VTOL cruise.

ASK: WHICH ONE SUITS A QUADCOPTER?

Worked example: a 16 kg spray drone chooses a series hybrid

Mission: 16 kg liquid payload, hover thrust 360 N (MTOM ≈ 36.7 kg), 20 min sortie, fast field turnaround.

Pure electric (LiPo)

- P = 36,700 g ÷ 6.0 g/W = 6,117 W
- E = 6.117 kW × 0.40 h = 2.45 kWh
- (20 min + 20% reserve)
- Battery = 2,450 ÷ 180 = 13.6 kg
- Recharge needs a field generator anyway

Series hybrid

- Generator 6.5 kW, 8.5 kg dry
- Fuel = 6.5 kW × 0.45 kg/kWh = 2.92 kg/h
- 20 min → 0.97 kg fuel
- Generator + fuel = 9.47 kg
- Refuel in about 30 s

Verdict: 4.1 kg lighter and a 30-second turnaround. The cost is vibration, noise and engine maintenance.

A bigger, slower propeller hovers on less power

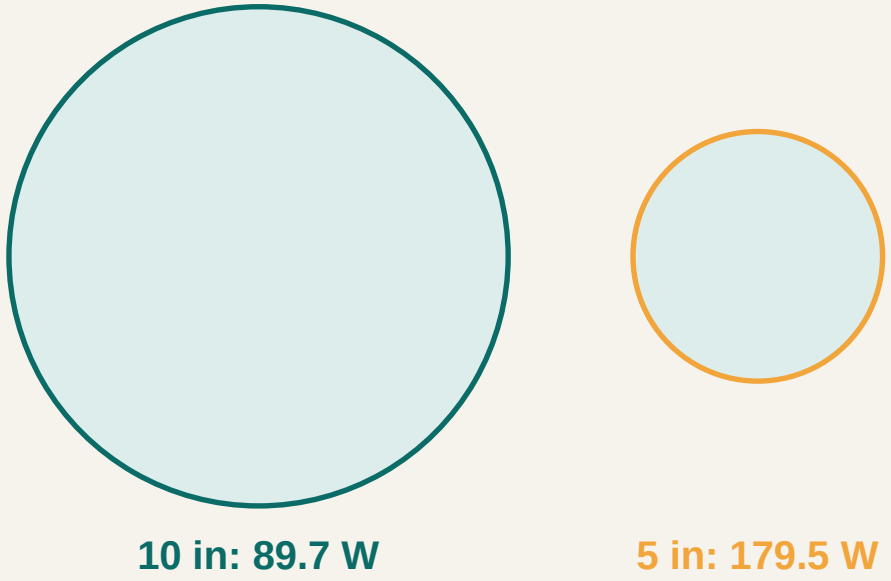
Momentum theory, ideal hover power

$$P = T^{1.5} \div \sqrt{(2\rho A)}$$

$A = \pi D^2/4$, $\rho = 1.225$ kg/m³

Rotor (T = 10 N)	Disk area	Ideal power
D = 254 mm (10 in)	0.0507 m²	89.7 W
D = 127 mm (5 in)	0.0127 m²	179.5 W

Halve the diameter and you need √4 = 2× the power for the same thrust.



Scaling laws at fixed propeller shape:

$T \propto n^2D^4 \quad P \propto n^3D^5$

Big props respond slowly, so racing quads use small ones.

ASK: THEN WHY NOT FIT HUGE PROPS?

Pick the sensor for the question, then fly for the resolution

RGB

Maps, photogrammetry, visual defects

Multispectral

Crop health (NDVI), water stress

Thermal (radiometric)

Hot spots, solar cells, leaks, search

LiDAR

Terrain under vegetation, power-line clearance

Ground sampling distance

GSD = pixel pitch × height ÷ focal length

p = 2.4 µm, H = 100 m, F = 8.8 mm

GSD = $2.4\times10^{-6} \times 100 \div 8.8\times10^{-3}$

= 0.027 m ≈ 2.7 cm per pixel

Your turn: what height gives 1.5 cm/px with the same camera?

RTK corrects in flight; PPK corrects after landing

RTK: real-time kinematic

- Base station sends corrections (RTCM) by radio or NTRIP
- 1–2 cm while the link holds
- Position known during flight: precise landing, docking
- Link loss = fall back to meter-level GNSS

PPK: post-processed kinematic

- Drone and base both log raw GNSS (RINEX)
- Processed on a laptop after the flight
- No link needed, so no dropouts in the geotags
- Accuracy not available in flight

Both remove most ground control points. Keep a few checkpoints to prove the accuracy.

ASK: WHICH SURVIVES A RADIO DROPOUT?

Match the job to the drone, sensor and positioning

1	**Paddy-field crop stress survey (50 ha)**	A	Fixed-wing VTOL + payload bay + BVLOS link
2	**Solar-farm hot-cell inspection**	B	Multirotor + radiometric thermal
3	**Stockpile volume at a quarry**	C	Fixed-wing + multispectral
4	**Blood sample delivery, 18 km**	D	Multirotor + RGB + RTK/PPK
5	**Power-line vegetation clearance**	E	VTOL + LiDAR + PPK

Then add one reason per line: endurance, accuracy, or the physics the sensor measures.

NDVI turns two bands into a crop-health number

NDVI = (NIR − Red) ÷ (NIR + Red)

Range −1 to +1. Healthy leaves reflect NIR strongly and absorb red.

Pixel	NIR	Red	NDVI	Reading
Healthy rice	0.50	0.08	0.72	Dense, vigorous
Stressed patch	0.30	0.15	0.33	Scout this area
Bare soil (try)	0.25	0.20	?	?



≈ 0 ≈ 0.8

Stress map → prescription file → spray drone doses only where needed.

PANEL: COMPUTE BOTH PIXELS

Drones pay off through four value drivers

Cost	Time	Safety	Data quality
Fewer crew hours, no helicopter, no scaffolding	Hours instead of days; faster decisions	Nobody climbs the tower or enters the confined space	Repeatable, geotagged, measurable records

A business case quantifies at least one driver in rupees and states what it assumes.

ASK: WHICH DRIVER MATTERS MOST FOR A BRIDGE?

Payback = capital ÷ annual savings

Illustrative example: tower inspection, one district

Item	₹ lakh
Drone kit, software, training (capital)	12
Manual inspection cost per year	30
Drone operating cost per year	8
Annual savings (30 − 8)	22

Always state what the formula subtracts. Here ROI uses net gain after the capital spend in year one.

Result

Payback = 12 ÷ 22 = 0.55 yr ≈ 6.5 months

First-year ROI = (22 − 12) ÷ 12 = 83%

Your turn: operating cost rises to ₹14 lakh. New payback and ROI?

EXIT TICKET

One mission. One power source. One number.

1. **Pick electric or hybrid for a 3-hour pipeline patrol, and give one reason.**

2. **Compute the GSD for p = 3.3 µm, H = 80 m, F = 12 mm.**

3. **Name the value driver that justifies a drone for chimney inspection.**

Write directly on the panel or answer in 30 seconds with your partner.

CO3 - Apply